prevents us from understanding and valuing other animals on their terms.

In the early 1900s, a horse called Clever Hans became a German celebrity. Touring Germany's towns and villages, Hans showed off a remarkable grasp of the German language, and an even more remarkable mastery of mathematics. When asked, 'Hans, what is four times three?' Hans tapped his hoof twelve times. When shown a written message asking, 'What is twenty minus eleven?' Hans tapped nine times, with commendable Prussian precision.

In 1904 the German board of education appointed a special scientific commission headed by a psychologist to look into the matter. The thirteen members of the commission – which included a circus manager and a veterinarian – were convinced this must be a scam, but despite their best efforts they couldn't uncover any fraud or subterfuge. Even when Hans was separated from his owner, and complete strangers presented him with the questions, Hans still got most of the answers right.

In 1907 the psychologist Oskar Pfungst began another investigation that finally revealed the truth. It turned out that Hans got the answers right by carefully observing the body language and facial expressions of his interlocutors. When Hans was asked what is four times three, he knew from past experience that the human was expecting him to tap his hoof a given number of times. He began tapping, while closely monitoring the human. As Hans approached the correct number of taps the human became more and more tense, and when Hans tapped the right number, the tension reached its peak. Hans knew how to recognise this by the human's body posture and the look on the human's face. He then stopped tapping, and watched how tension was replaced by amazement or laughter. Hans knew he had got it right.

Clever Hans is often given as an example of the way humans erroneously humanise animals, ascribing to them far more amazing abilities than they actually possess. In fact, however, the lesson is just the opposite. The story demonstrates that by humanising animals we usually *underestimate* animal cognition and ignore the unique abilities of other creatures. As far as maths goes, Hans was hardly a genius. Any eight-year-old kid could do